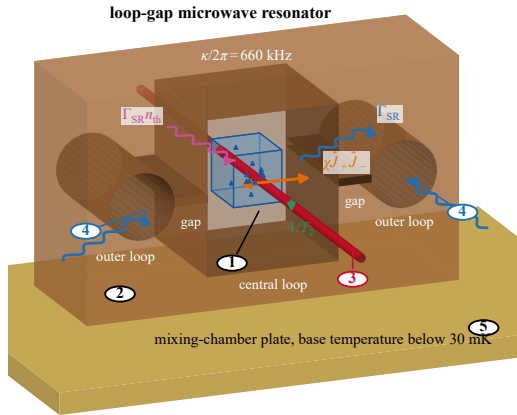
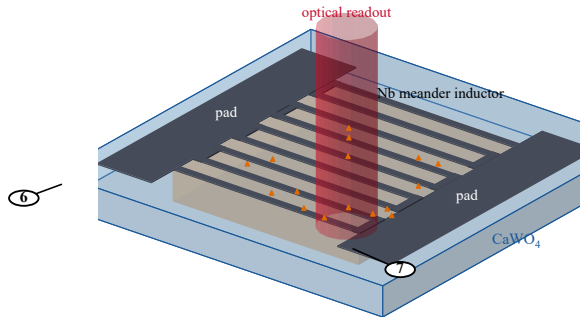


(a)**Key**

- ① $^{171}\text{Yb}^{3+}:\text{CaWO}_4$ crystal, $4.4 \times 4.6 \times 5 \text{ mm}$, 4.96 ppm; c axis along the beam
- ② loop-gap resonator: central loop, two outer loops, gaps; $g/2\pi = 15 \text{ mHz}$, $V_m = 275 \text{ mm}^3$
- ③ 973 nm optical input and output: pumping into $|\downarrow\rangle$, absorption readout
- ④ microwave antennas at the outer loops: pulses, transmission S_{21}
- ⑤ mixing-chamber plate; spin temperature 80 mK, $n_{\text{th}} = 0.19$
- ⑥ CaWO_4 chip with a Nb meander inductor between capacitor pads
- ⑦ read-out volume under the inductor (near-uniform field, coupling spread D)

(b) planar superconducting resonator on the same crystal (design-map case, proposed, not built)

$$\kappa/2\pi = 3 \text{ to } 300 \text{ kHz}, g\sqrt{N}/2\pi = 0.1 \text{ to } 5 \text{ MHz}, N \approx 10^9 \text{ to } 10^{11}, 20 \text{ mK}$$

(c)